Cold Starts in Managed Vector Search: A Two-Week Measurement

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CS 6100: Information Retrieval

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October 20, 2026

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Introduction

Managed vector indexes are memory resident, so an index that has been scaled down must be reloaded before it can answer. Operators report this as an occasional slow query; for a person opening a search box it is the first impression of the system.

Published evaluations usually report warm percentiles, because warm measurement is easy to automate (Example & Sample, 2025). This report asks how large the cold-start penalty is in practice and what it scales with.

Method

One query was issued every fifteen minutes for fourteen days against a three-replica index of 4.2 million 768-dimensional vectors. A query counted as cold when the preceding idle gap exceeded ten minutes. Warm queries ran in a separate stream against the same index, so both distributions come from the same days.

Measures

Client time to first byte and the service's own reported latency were recorded for every query, together with the replica that answered and the index size at that hour.

Results

The cold median was 780 ms against a warm median of 110 ms, a factor of 7.1. The gap widened in the tail: the cold 90th percentile was 4,100 ms against 190 ms warm. Cold starts were 3.1% of queries but 41% of the time spent waiting.

Cold-start latency correlated with index size (r = .71) and not with replica count (r = .08), which points at segment loading rather than request routing. Raising the minimum replica count

from one to two and enabling the keep-warm option removed 86% of cold starts above one second, at 19% more instance hours.

Discussion

The measurement covers one service, one index shape and one region, so the numbers are not a ranking. What generalises is the shape of the distribution: a heavy tail that steady-state benchmarks do not show and that a keep-warm setting can remove.

References

Example, A., & Sample, B. (2025). A survey of approximate nearest-neighbour benchmarks. Journal of Invented Systems, 12(3), 210-219. https://doi.org/10.0000/jis.2025.12.3.210

Placeholder, C. (2026). Notes on index reload behaviour in shared clusters. In Proceedings of the Invented Conference on Systems Measurement (pp. 44-51). Invented Press. https://doi.org/10.0000/icsm.2026.44